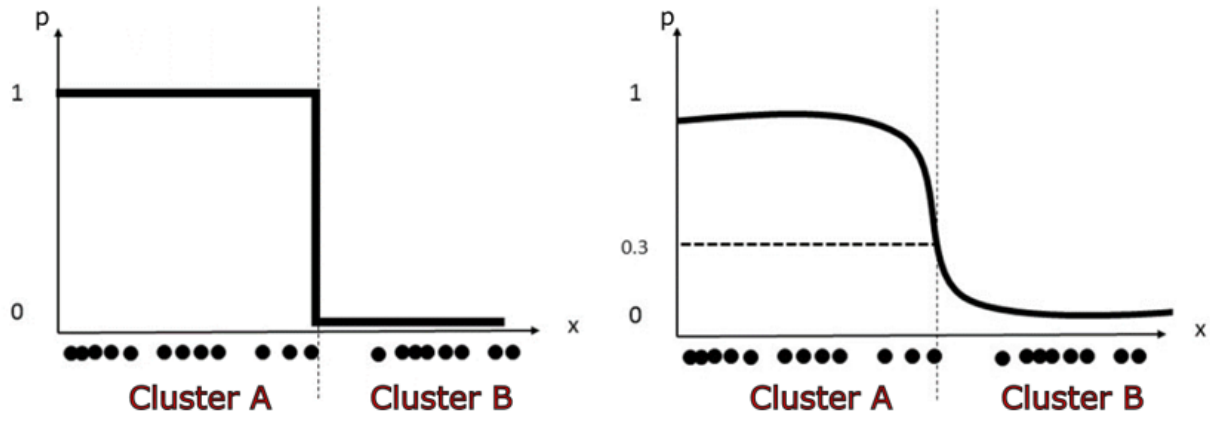


Figure 12.2 The difference between hard and soft clustering can be explained in terms of the membership function they adopt: hard clustering's outputs either `0` or `1` for each point and each cluster, with the constraint that it can be `1` for only one of the point-cluster combinations; soft clustering's membership function, instead, outputs a probability between `0` and `1` (any value in-between) and for each points, it can be non-zero for multiple clusters.

A first relevant distinction is made between hard and soft clustering, as also shown in figure 12.2.

- In **hard clustering**, to every point the output assigns a single cluster, one and only one (or at most one, if the clustering algorithm can also detect noise).
- In **soft clustering**, instead, for each point `P` and each group `G`, the output provides a probability that `P` belongs to `G`.

The other main criterion to classify clustering algorithms differentiates *partitioning clustering* from *hierarchical clustering*: